

EFFECT OF ADRENALINE AND ATP ON GLYCERINATED SKELETAL MUSCLE

MOHAMMED SALEH WAHEEB MAHMOUD

Department of Physiology,
Faculty of Medicine and Medical Sciences
Umm-al-Qura University, Makkah, Saudi Arabia

ABSTRACT:

Glycerinated psoas muscles of rabbit were used to study the effect of adrenaline and ATP on the contractile machinery of skeletal muscle. The muscles were placed in pure glycerine for 22 days to rupture all the membranes and then a few fibres were stripped off. The tension was recorded with the help of a highly sensitive force transducer in the presence of adrenaline and ATP. The results obtained showed a 19% higher tension generated in the presence of adrenaline and this increase in tension was significant statistically. Since the muscles used were devoid of their membranes and electrical potentials, it was assumed that this drug produced higher tension by acting directly on the contractile proteins themselves.

Key words: Adrenaline, ATP, glycerinated muscle, contractile proteins, tension.

INTRODUCTION

It has been suggested that adrenaline secreted by splanchnic stimulation has the same effect on non-fatigued and fatigued skeletal muscles and that it can not be regarded as a specific antagonist to fatigue substances (Gruber, 1922, Axelsson, 1961, Garcia, 1994). There are also evidences that sympathetic stimulation or the application of adrenaline and nor-adrenaline on the intact and isolated muscles increased the twitch tension (Cairns, 1993, Kissling, 1977). Similarly, Bueding and Bulbring (1964) have found that adrenaline increased the twitch tension in frog rectus abdominus muscle and that this effect was more prominent in the muscles of adrenalectomized animals.

It has also been shown that adrenaline increased the height of contraction and irritability of the muscle but no change was observed in the electrical excitability of these muscles after the administration of adrenaline (Forbes, 1980, Hammarstrom, 1997). According to these authors, the defatiguing effect of adrenaline was probably due to a decrease in the threshold to acetylcholine at the myoneural junction. Bulbring (1957), Jonsdottir (1996) have also suggested that the defatiguing effect of adrenaline was due to blood pressure changes, but Stassen, (1997) doubted whether the effect of adrenaline on blood vessels in the active muscles was associated with fatigue removal.

In the recent studies, Savino, (1996), Layland (1997) observed the effect of

intravenous perfusion of adrenaline on hyperlactacidaemia in humans and animals and concluded that it was impossible to decide on a real physiological meaning of hyperlactacidaemic effect.

There are thus, clear indications that adrenaline produces some effect on the actively contracting skeletal muscles. However, it remains to be determined whether adrenaline produced its effects by acting on the receptors located on the surface of muscle membrane or by acting directly on the contractile proteins, actin and myosin or by some other means. A direct effect of adrenaline on the contractile proteins can only be possible provided this drug can diffuse and reach them.

The present work is therefore, undertaken to study whether adrenaline directly affect the contractile proteins to increase tension generation in the glycerinated muscles.

MATERIALS AND METHODS

Rabbit psoas muscles were dissected out and placed in 100% glycerine for 21 to 22 days in a stretched position by a technique described previously (Larsson, 1995). Since the cell membrane is not a barrier to the influx of glycerine, it enters the cell and ruptures all the membranes within this time period. After 22 days, a few fibre bundles were removed and washed thoroughly with a solution containing 117 mM KCl, 20 mM NaCl and 10 mM ATP. The fibres were then placed in a muscle chamber, soaked in the same medium described above and were attached to a very sensitive force transducer having a maximum sensitivity of 1 gram. For tension recordings, a few drops of 2 mM CaCl_2 were added to the medium and the contracture developed was recorded using the Bioscience MD2 Oscellograph. Whenever, effect of adrenaline was to be tested, the muscle fibres were soaked in adrenaline 10^{-6} for about 2-5 minutes prior to their fixation in the muscle chamber for the recording of tensions. The muscle fibre bundles were then weighed and their lengths were measured. The tension produced by the muscle fibre bundles were then calculated in units of mg/cm^2 .

RESULTS

The results obtained from 9 different muscle fibre bundles are shown in Table-1. The tension produced by the muscle fibre bundles in the presence of ATP, averaged to about $24.23 \pm 1.74 \text{ mg}/\text{cm}^2$ while the tension generated in the presence of adrenaline plus ATP averaged to about $28.78 \pm 1.61 \text{ mg}/\text{cm}^2$. These results clearly demonstrated that the presence of adrenaline has increased the tension by about 19.25%. This increase in tension was significant statistically ($P < 0.005$).

DISCUSSION

Oliver and Schafer (1895) were the first to demonstrate that extracts of adrenal medulla increased the tension and the duration of twitch in the dog gastrocnemius muscles. Later, a number of workers have confirmed that adrenaline, and to a lesser extent, nor-adrenaline markedly increased both the twitch and tetanic tensions of fatigued slow and fast contracting mammalian muscles (Gruber, 1917, 1924; Brown et al., 1947, 1948; Parry and Squire 1973, Gleeson et al., 1980). However, Katz (1967), Nakamaru et al. (1972) found no effect of nor-adrenaline on the properties of cardiac myosin and actomyosin filaments. Other investigations have however, shown an increase in the twitch tension and an increase in total duration of twitch (Goffart and Ritchie, 1952; Boaman and Zaimis, 1958; Penny and Goldspink, 1979; Eltze, 1996; Kossler et al., 1997).

Table 1
Effect of adrenaline on the glycerinated psoas muscle of rabbit
The muscles were glycerinated for 21 to 22 days for rupturing
the muscle membranes. Adrenaline was used in 10^{-6} concentrations

No .of Fibre Bundle	Weight of the Fibre (mg)	Length of the Fibre (cm)	ATP	Adrenaline + ATP	Tension Produced (mg)	Tension Produced (mg/cm^2)
1	3.0	0.8	-	+	20.1	5.4
2	1.5	0.7	-	+	23.4	10.9
3	1.0	1.5	+	-	12.4	19.1
4	1.0	2.0	+	-	30.4	60.9
5	0.7	0.5	-	+	3.8	22.7
6	0.7	0.7	-	+	20.1	20.1
7	1.5	1.1	+	-	58.2	42.7
8	1.5	1.6	-	+	46.8	49.6
9	1.0	1.3	-	+	11.4	14.8

Tension (mg/cm^2) with ATP only = 24.21 ± 1.73

Tension (mg/cm^2) with ATP + Adrenaline = 28.78 ± 1.61

However, our results have demonstrated a 19% increase in tension when glycerinated muscles were soaked in adrenaline plus ATP medium. Since glycerination of the muscle ruptures all the muscle membranes, it is quite possible that adrenaline has diffused to the contractile proteins to increase the tensions. Our results

are thus in conformity with those of Goffart and Ritchie (1952); Urwyler (1993); Moreau et al. (1995) who have shown that the initial increase in twitch tension was also associated with a prolongation in the duration of the active state plateau. A similar prolongation in the duration of active state plateau in twitch has also been reported recently (Silva et al., 1996; Hammarstrom, 1997).

The present findings thus provide evidence that adrenaline has increased the tension in our experiments by acting directly on the contractile proteins in general and probably on myosin in particular in our opinion this became possible since the muscle membranes were ruptured by glycerination and adrenaline had no difficulty in diffusing through the muscle mass to reach the level of contractile proteins. In addition, we also conclude that adrenaline increases the twitch tensions by prolonging the duration of active state, probably by releasing more free ionic calcium from its binding sites within the muscle fibres.

REFERENCES

- Axelsson J., Bueding E., Bulbring (1961). *J. Physiol. (Lond.)* 156: 357.
 Brown W.C., Zaimis E. (1958). *J. Physiol.* 144:92.
 Brown G.L., Bulbring E., Burns B.D. (1947). *J. Physiol.* 106: 15.
 Brown G.L., Bulbring E., Burns B.D. (1948). *J. Physiol.* 107: 115.
 Bueding E., Bulbring E. (1964). The inhibitory action of adrenaline. Biochemical and biophysical observations. In: "Pharmacology of smooth muscles" pp.37-57. Ed. E. Bulbring Pergamon Press Inc., Oxford.
 Bulbring E., Burn J.H. (1939). *J. Physiol.* 95: 203.
 Bulbring E. (1957). *J. Physiol. (Lond.)*, 135: 412.
 Cairns S.P., Dulhunty A.F. (1993). *Br. J. Pharmacol.* 110(3): 1133.
 Carslen R.C., Gray S.D., Picker J.G. (1996). *Acta. Physiol. Scand.* 156: 237.
 Eltze M. (1996). *Eur. J. Pharmacol.* 311 (2-3): 187.
 Forbes M.S. and Sperelakis N. (1980). *J. Ult. Res.* 73: 254.
 Garcia M.C., Escamilla S.J. (1994). *Can. J. Physiol. Pharmacol.* 72(12): 1580.
 Gleeson T.T., Putnam R.W., Bennet A.F. (1980). *J. Exp. Zool.* 214(3): 293.
 Goffart M., Ritchie J.M. (1952). *J. Physiol.* 116: 357.
 Gruber C.M. (1917). *Am. J. Physiol.* 62: 438.
 Gruber C.M. (1922). *Am. J. Physiol.* 62: 530.
 Gruber C.M. (1924). *J. Pharmacol. Expt. Ther.* 23: 325.
 Hammarstrom L.E., Andersson K.E., Holmin T. (1997). *Res. Exp. Med. Berl.* 196(6): 373.
 Jonsdottir I.H., Johansson C., Hoffman P. (1996). *Acta. Physiol. Scand.* 157: 443.
 Juchmes J., Fossio A.C., Sheen A. (1973). *Arch. Int. Physiol. Biochem.* 81: 783.
 Katz B. (1967). *Am. J. Physiol.* 212: 1.

- Kissling G., Blickle B., Ross C., Gulbines E. (1997). *J. Physiol. (Lond.)* 499: 195.
- Kossler F., Nasodev G.A., Shvinka N. (1996). *Gen. Physiol. Biophys.* 15(1): 37.
- Larsson S.E., Larsson R., Oberg P.A. (1995). *Eur. J. Appl. Physiol.* 71: 493.
- Layland J., Young I.S., Altringham J.D. (1997). *J. Exp. Biol.* 200: 503.
- Moreaue D., Dubots P., Boggio V., Guillard J.C., Cometti G. (1995). *J. Sports. Sci.* 13(2): 95.
- Nakamaru Y., Schwartz A. (1972). *J. Gen. Physiol.* 59: 22.
- Oliver G., Schafer E.A. (1995). *J. Physiol.* 18: 230.
- Parry D.A., Squire J.M. (1973). *J. Mol. Biol.* 75: 33.
- Penny R.K., Goldspink G. (1979). *J. Univ. Kuwait (Sci)* 6: 159.
- Savino E.A., Varela A., Carregal M. (1996). *Rev. Esp. Fisiol.* 52: 155.
- Silva E.G., Vianna L.M., Paira T.B. (1996). *Br. J. Pharmacol.* 118: 1367.
- Stassen F.R., Willemsen M.J., Janssen G.M., Smit J.F., Mey J.G. (1997). *Br. J. Pharmacol.* 120(8): 1505.
- Urwyler A., Censier K., Kaufman M.A., Drewe J. (1993). *Br. J. Anaesth.* 70(1): 76.